Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY 8875/02

Paper 2 Structured and Free-Response Questions

16 September 2013

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section B Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section C Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided.

Submit your answers to Sections B and C separately.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

For Examiner's Use	
Section A [30]	
Section B [40]	
1	/ 10
2	/ 5
3	/ 7
4	/ 6
5	/ 12
Section C [20]	
1 / 2	
Total [90]	

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For Examiner's use

Section B: Structured Questions (40 marks)

Answer **all** questions in this section.

Question 1

(a) **Fig 1.1** shows a section of a cell.

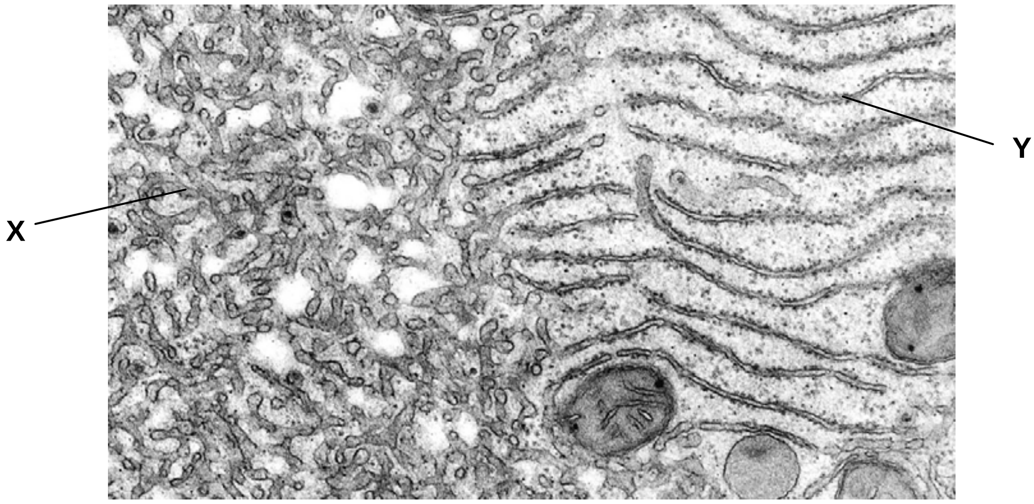


Fig 1.1

Name organelles **X** and **Y**, and explain their functions. [5]

X:

Y:

(b) **Fig 1.2** shows diagrams which represent lipid and lipid-related molecules.

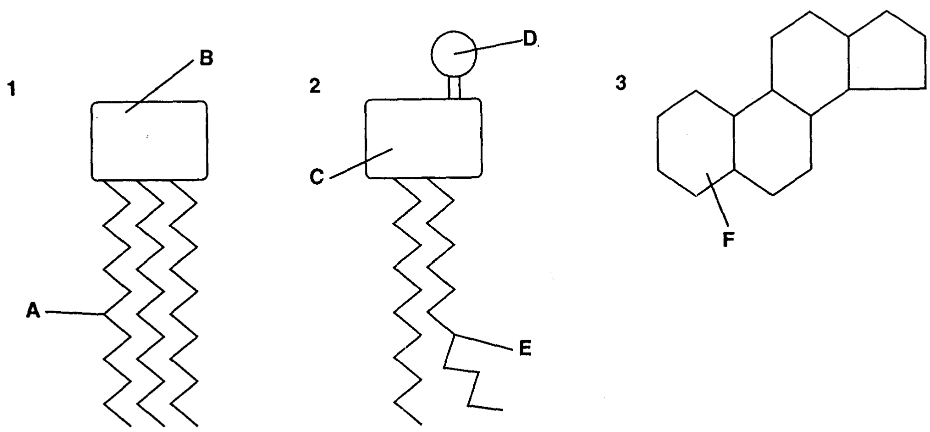


Fig 1.2

(i) Explain the function of molecules **2** and **3** in membranes. [4]

Molecule **2**:

Molecule **3**:

(ii) Which one of the components, labelled **A** to **F** in **Fig 1.2**, would most likely be metabolised in order to provide energy if a person goes on a low calorie diet? [1]

Total: [10]

Question 2

For Examiner's use

In Drosophila flies, the wing shape and eye colour are controlled by two different genes found on separate chromosomes. The gene controlling wing shape is found on chromosome 2 and has 2 alleles. Curly wings were found to be dominant over normal wings. **Fig 2.1** and **Fig 2.2** shows the normal wing and curly wing phenotype respectively.

Fig 2.1: Normal Wing

Fig 2.2: Curly Wing

The gene controlling eye colour is found on the X chromosome and has two alleles. It was observed that red eye trait is dominant over white eye trait.

(a) Explain what is meant by the term "***dominant***"? [1]

(b) Construct a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when a white eyed female is crossed with a red eyed male, both are heterozygous for curly wings. [4]

Total: [5]

Question 3

For Examiner's use

In an investigation, a student A prepared mixtures using different concentrations of leaf extracts and DCPIP solution. To measure the rate of photosynthesis, he had to record the time it takes for the mixture to change colour from blue-green to green.

Table 3.1 shows his result.

<u>Table showing the effects of different concentrations of leaf extract / % on the time taken for DCPIP to decolourise /s and the rate of reaction / s^{-1}</u>

Concentration of leaf extract / %	Time taken for the dye to decolourise / s	Rate of reaction (1/t) / s^{-1}
100	90	
80	120	
64	210	
51	330	

Table 3.1

(a) Complete the table with the rate of reaction for all concentrations of leaf extract. [2]

(b) Plot the rate of reaction against the different concentrations of leaf extract on the following page. [3]

(c) At the end of this experiment, Student A concluded that the decolorisation of DCPIP was due to photosynthesis.

Student B begged to defer and conducted another experiment to prove Student A wrong. She placed the same experimental setup in total darkness for the same duration and found that the DCPIP decolorized too. Explain her observation. [2]

Total: [7]

For Examiner's use

Question 4

(a) In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

Fig 4.1 shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.

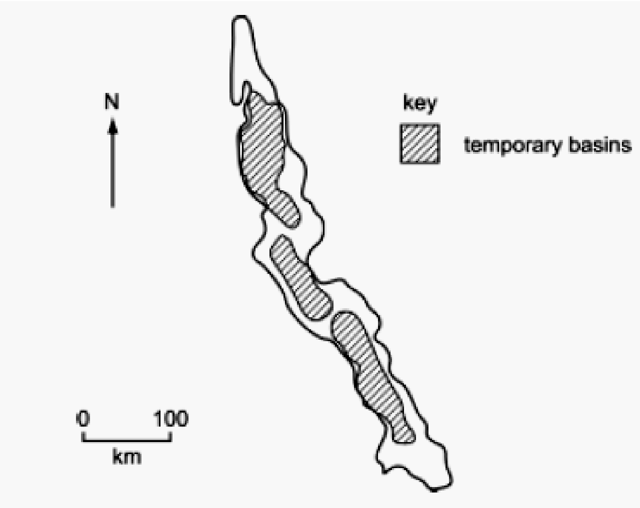


Fig 4.1

Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [4]

(b) Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus*. **Fig 4.2** shows part of the cytochrome b DNA sequence.

T. brichardi	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
T. kasabae	CTTTTAACTATGATAATAGCTTTTGTGGGCTACG
T. annectens	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
T. moorii	CTTTTAACTATGATAACAGCTTTTGTGGGTTACG
T. polli	CTTTTAACTATAATAACAGCTTTTGTGGGCTACG
T. duboisi	CTTTTAACCATGATAACGGCTTTTGTAGGCTATG

Fig 4.2

With reference to **Fig 4.2**, explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]

Total: [6]

Question 5

For Examiner's use

Fig 5.1 shows two plasmids pAMP and pKAN. Amp[r] and Kan[r] confer resistance to the antibiotics ampicillin and kanamycin respectively.

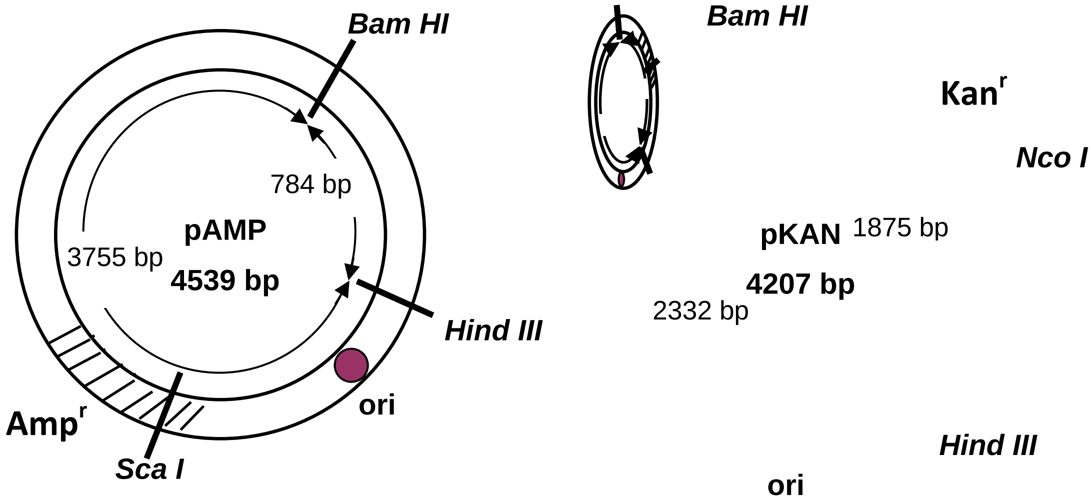


Fig 5.1

In an experiment to make a cloning vector, solution containing the plasmids pAMP and pKAN were treated with a solution containing restriction enzymes BamHI and HindIII. DNA ligase was then added and the mixture was incubated at 37°C.

This experiment produced the plasmid pAK which can be used as a cloning vector. The restriction map of pAK is shown in **Fig 5.2** and the specific recognition sequence of its restriction sites are shown in **Table 5.3**.

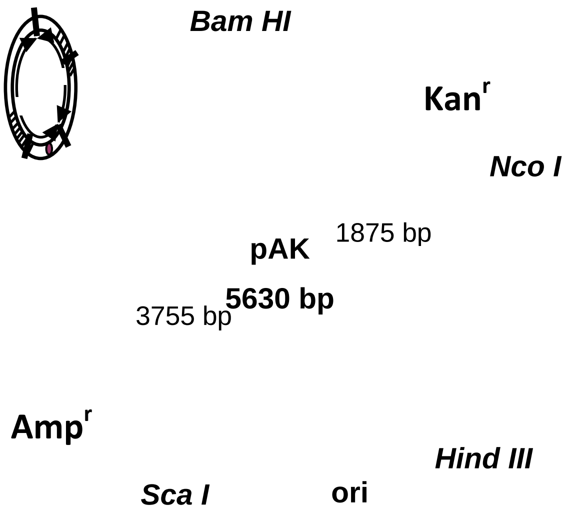


Fig 5.2

Restriction Enzyme	Recognition Sequence
Bam HI	5'-G^G A T C C-3' 3'-C C T A G^G-5'
Hind III	5'-A^A G C T T-3' 3'-T T C G A^A -5'
Sca I	5'-A G T^A C T-3' 3'-T C A^T G A-5'
Nco I	5'-C^C A T G G-3' 3'-G G T A C^C-5'

^ indicates where the restriction enzyme cuts

Table 5.3

(a) Restriction sites are palindromic. Explain the meaning of the term "palindromic". [1]

(b) With reference to **Fig 5.2** and **Table 5.3**, explain why Nco I is a more suitable choice of restriction enzyme for use in genetic engineering with the pAK plasmid than Sca I. [2]

To clone a human gene with a molecular size of 600 bp in bacterial cells, the following proceedure was carried out as shown in **Fig 5.4**.

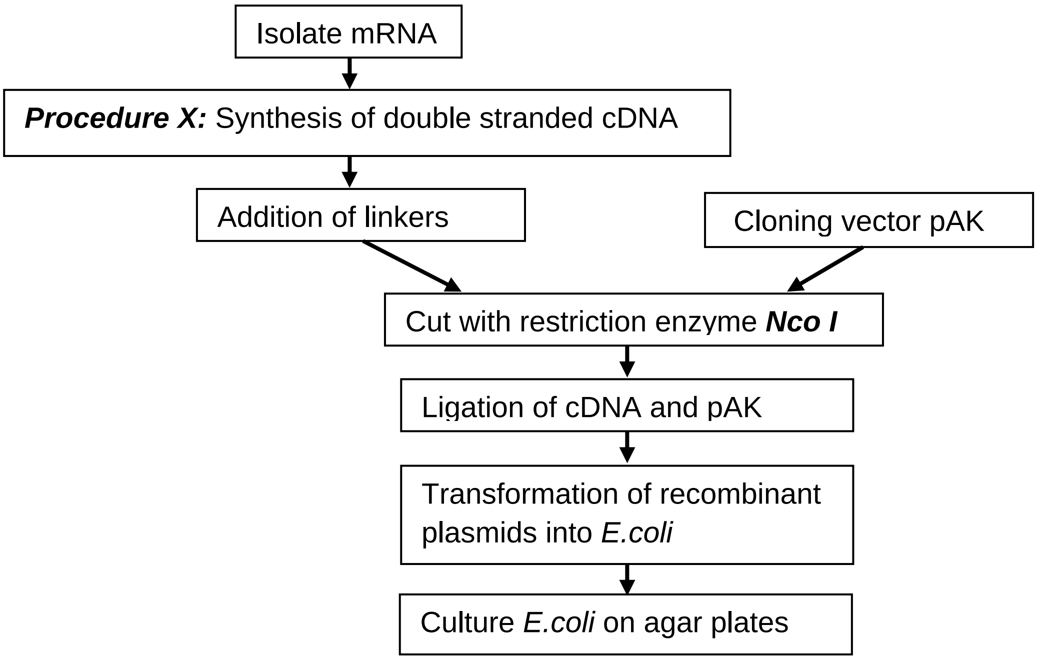


Fig 5.4

(c) With reference to **Fig 5.4**, briefly describe procedure X. [3]

(d) Draw **three** possible gene products formed in the ligation mixture. [3]

(e) Explain how the plasmid pAK can aid in the successful selection of recombinant colonies with the gene of interest. [3]

Total: [12]

Section C: Free-Response Questions (20 marks)

Answer only **one** question.
Write your answers on the writing paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 1

(a) Describe how you would carry out an investigation to determine the effect of substrate concentration on the rate of enzyme reaction, using hydrogen peroxide and potato catalase. [10]

(b) **(i)** Describe Krebs Cycle. [5]

(ii) Compare oxidative phosphorylation and photophosphorylation. [5]

OR

Question 2

(a) Describe the process of transcription of mRNA in eukaryotes. [10]

(b) **(i)** Describe how molecular methods can be used to classify organisms. [6]

(ii) Suggest how bacteria evolve. [4]

Total: [20]

END OF PAPER

DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2013
YEAR SIX
H1 BIOLOGY (8875)

Answer Scheme

Structured Answers

Question 1

(a)
X: smooth endoplasmic reticulum;
Synthesis of lipids / detoxification of drugs in liver / storage and release of calcium ions in muscles;

Y: rough endoplasmic reticulum;

Synthesis of proteins;
Folding and post-translational modification of proteins;
Transport of proteins to Golgi Apparatus;
2 max

(b)(i)
Molecule 2: phospholipid

Amphipathic, forms a bilayer with hydrophilic phosphate heads facing outwards towards aqueous medium and hydrophobic fatty acid tails facing inwards

Forming a hydrophobic barrier / acts as a barrier to polar / charged substances, thus contributing to the selective permeability of cell membranes

Molecule 3: cholesterol

Controls fluidity by increasing fluidity of cell membrane by disrupting the close and orderly packing of the phospholipids during low temperature

or

decreasing fluidity of cell membrane by conferring a stiffening and strengthening effect during high temperature;

Improves membrane impermeability by acting as a “mortar” that fills in small gaps in the phospholipid structure;

(ii)
A

Question 2
(a)
In a heterozygous individual, the phenotype of the dominant allele is always expressed and masks the effect of the recessive allele;

(b)

Let **A** represent the dominant allele for curly wings
a represent the recessive allele for normal wings
X^R represent the dominant allele for red eyes
X^r represent the recessive allele for white eyes

Parental phenotype:	White-eyed female with curly wings	X	Red-eyed male with curly wings
Parental genotype:	X^rX^rAa	X	X^RYAa
Gametes:	X^rA X^ra	X	X^RA X^Ra YA Ya

[1]

Punnets square:

	X^RA	X^Ra	YA	Ya
X^rA	**X^RX^rAA**	**X^RX^rAa**	**X^rYAA**	**X^rYAa**
X^ra	**X^RX^rAa**	**X^RX^raa**	**X^rYAa**	**X^rYaa**

[1]

Offspring genotype:	**X^RX^rAA** **2 X^RX^rAa**	**X^RX^raa**	**X^rYAA** **2 X^rYAa**	**X^rYaa**
Offspring phenotype:	Red eyed female with curly wing	Red eyed female with normal wing	White eyed male with curly wing	White eyed male with normal wing
Offspring phenotypic ratio:	**3**	**1**	**3**	**1**

[1]

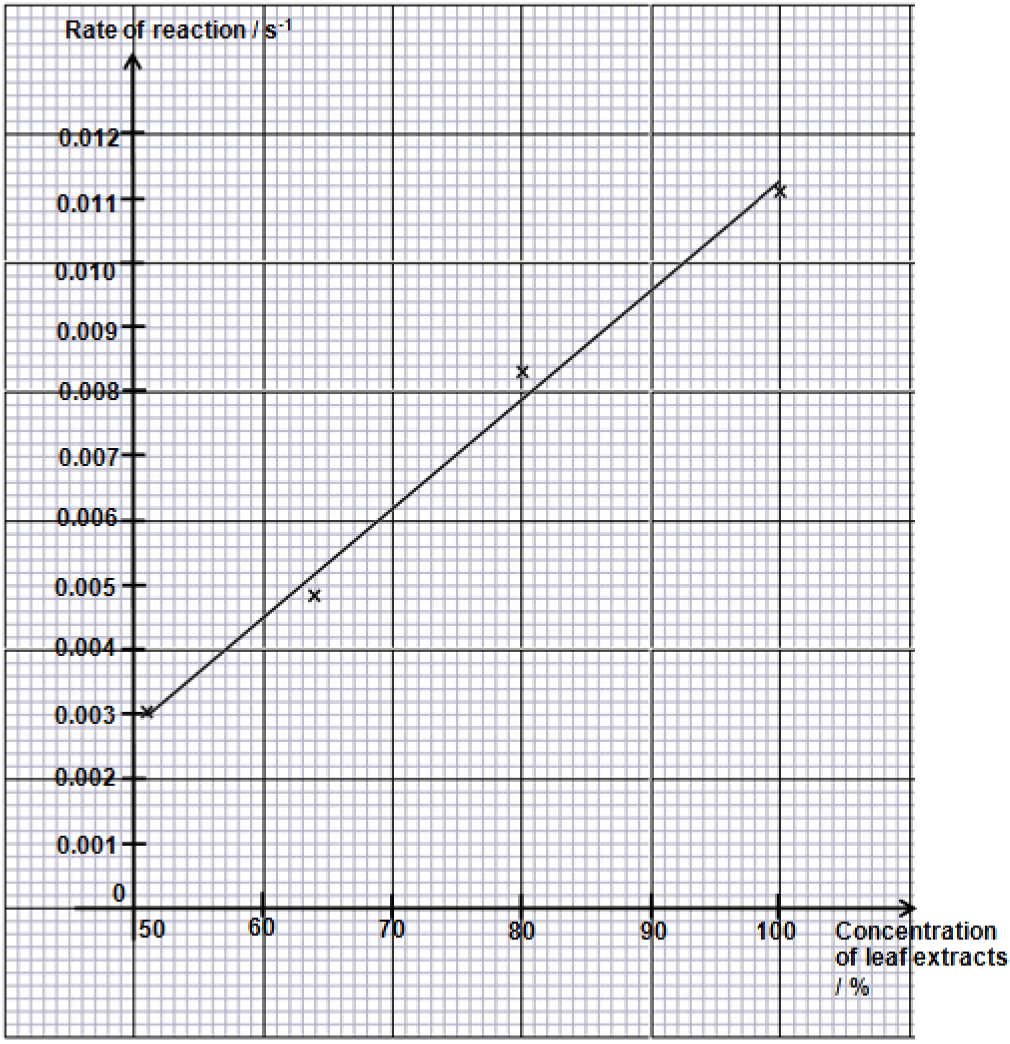
[1]

Question 3
(a)

Concentration of leaf extract / %	Time taken for the dye to decolourise / s	Rate of reaction (1/t) / s^{-1}
100	90	0.0111
80	120	0.0083
64	210	0.0048
51	330	0.0030

1 M for correct calculation
1 M for 3 s.f. or 4 d.p.

(b)



Label axes with units;
Correctly plot;
Best fit line;

(c)
Feature: dense region in the nucleoplasm or nucleus;
R: enclosed organelle / not bound by a membrane

Role: contains DNA that codes for rRNA / transcription of rRNA genes / formation of rRNA for ribosome synthesis;

(c)
Presence of mitochondria in leaf extract has the same effect as chloroplast;
DCPIP remove electron from the ETC of oxidative phosphorylation in respiration;

Question 4
(a) (i)
variations in the Tropheus fish population due to mutation resulting in different alleles;
at least 6 different niches in the lake with different selection pressure;
fish with at selective advantage survive and reproduce viable offspring;
passing on advantageous genes/alleles to the next generation;
accumulation of many genetic changes over a long period of time to evolve into 6 species;
reproductive/ behavioural/ sympatric/ allopatric/ geographical isolation/ accept hundreds of km apart thus no gene flow between different populations;
4 max

(b)
evolved from a common ancestor;
cytochrome b gene code for important protein involved in respiration thus any mutation will result in non-functional protein which will be fatal to the organism;
this gene would have gone through the least mutation not counting silent mutation;
3 max

Question 5
(a)
nucleotide sequences that are the same on both strands of the DNA molecule when read in the same direction (5' to 3') / read the same in the 5' to 3' direction on each strand of DNA.

(b)
Nco I produces staggered / sticky ends whereas Sca I produces blunt ends;

This facilitates the annealing of the gene of interest to plasmid when the plasmid and gene are cut with Nco I by complementary base pairing;

Using Sca I would require the additional procedure of adding linker DNA to generate sticky ends to facilitate annealing of gene of interest.

(c)

Oligo-dT primers anneal to 3' poly-A tail of mRNA

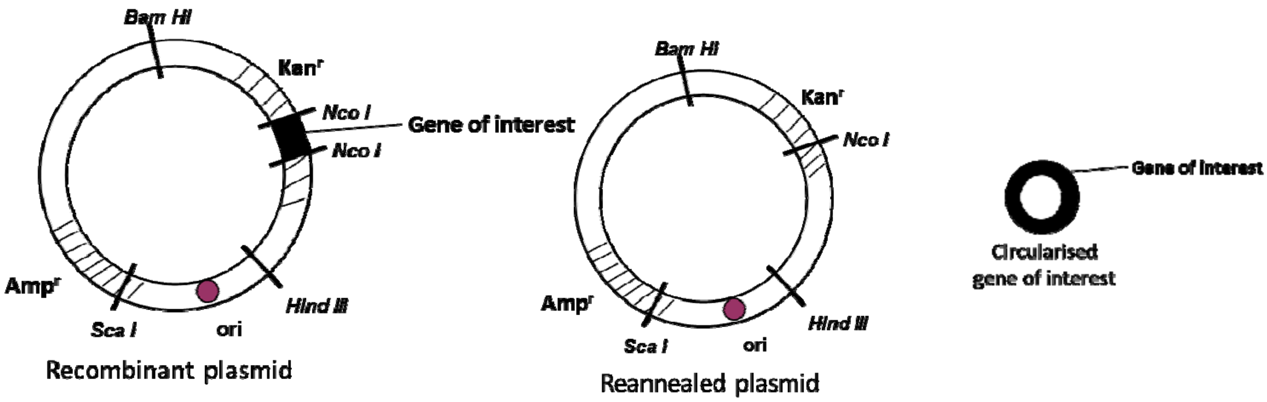
Reverse transcriptase extends primers using mRNA as template to synthesize single stranded cDNA, product is RNA-DNA duplex.

Partial digestion of mRNA using alkaline solution / RNase H

ref to synthesis of 2nd complementary strand to form double stranded cDNA using DNA polymerase

<u>3 max.</u>

(d)



1 mark each

(e)

Amp^R gene codes for ampicillin resistance thus allows for first selection for successfully transformed colonies which have taken in the plasmid;

Colonies with reannealed plasmids will have intact Kan^r gene which codes for kanamycin resistance and will survive on kanamycin plate;

Colonies with recombinant plasmids will gene of interest inserted into Kan^r gene wich results in insertional inactivation, and will not survive on kanamycin plate;

Essay Answers

Question 1
(a)

1. theory of how [S] affects rate

2. catalase reaction

3. measure rate of oxygen production to measure rate or reaction

4. predicted results

5. independent variable – 5 conc. of H_2O_2

6. dependent variable – measure of no of bubbles / vol of oxygen per unit time

7. control of other variables for enzyme reaction – temperature, pH, [enzyme]

8. dilution method

9. method to prepare potato

10. labeled and annotated figure

11. how to measure rate of oxygen production

12. replicates / repeats

13. table

14. graph

1 (b)(i)

1. Acetyl CoA and oxaloacetate undergo a condensation reaction. Acetyl CoA adds its two-carbon acetyl group to oxaloacetate to form **citrate**. Enzyme = citrate synthetase;

2. Citrate is isomerised to **isocitrate**. Enzyme = aconitase;

3. Isocitrate undergoes oxidative decarboxylation to form α**-ketoglutarate**. **Carbon dioxide** is lost in this process. NAD^+ is the hydrogen acceptor and **NADH** is formed. Enzyme = isocitrate dehydrogenase

4. α-ketoglutarate undergoes oxidative decarboxylation to form **succinyl CoA**. **Carbon dioxide** is lost in this process. NAD^+ is the hydrogen acceptor and **NADH** is formed. Enzyme = α-ketoglutarate dehydrogenase

5. Succinyl CoA is converted to **succinate**. **ATP** is formed by substrate level phosphorylation. Enzyme = succinyl CoA synthetase

6. Succinate undergoes dehydrogenation to form **fumarate**. FAD is the hydrogen acceptor and **$FADH_2$** is formed. Enzyme = succinate dehydrogenase

7. Fumarate undergoes hydration to form **malate**. Enzyme = fumarase

8. Malate undergoes dehydrogenation to form **oxaloacetate**. NAD^+ is the hydrogen acceptor and **NADH** is formed. Enzyme = malate dehydrogenase

1(b) (ii)
Similarities
Both processes synthesizes ATP by chemiosmosis, where energy released by the transport of electrons down electron transport chain is used to pump H+ against a concentration gradient. And as the H+ diffuses back down a concentration gradient, the energy released is used to phosphorylate ADP to ATP;
Both processes involves the transport of electron down ETC.
2 max

Differences

Features	Oxidative phosphorylation	Photophosphorylation	
Definition	Synthesis of ATP occurs as electrons are transferred from NADH or FADH2 to oxygen by an **electron transport chain**. The energy coupling occurs through a proton gradient.	Synthesis of ATP occurs as electrons are transferred from water and chlorophyll molecules to NADP by an **electron transport chain**. The energy coupling occurs through a proton gradient.	;
Location	On the inner membrane of mitochondrion.	on the thylakoid membrane	;
Reactions	Involves a series of redox reactions via a series of electron carriers in the electron transport chain.	hydrolysis of water molecule on top of redox reaction in the electron carriers.	;
Synthesis of ATP	by Oxidative phosphorylation	by both cyclic and noncyclic Photophosphorylation	;

3 max

Question 2
(a)

<u>Initiation of transcription</u>

1. **<u>Transcription factors</u>** search for and attach to the **<u>TATA box</u>** within the **<u>promoter</u>** region along the DNA.
2. **<u>RNA polymerase II</u>** then binds to the **promoter** with the aid of the transcription factors.
3. The completed assembly of transcription factors and RNA polymerase II bound to the promoter is called the **<u>transcription initiation complex</u>.**
4. Thus causing the **unwinding** of the DNA double helix at the **transcription** start site**.**

<u>Elongation of transcription</u>

5. The **anti-sense / template / coding strand** is used as a **template for transcription**.
6. RNA polymerase II begins synthesis of RNA by selecting **free ribonucleotides** complementary to the DNA template (ref to complementary base pairing rule: A pairing with T on DNA, U pairing with A on DNA, C pairing with G)
7. RNA polymerase II joins ribonucleotides together by **phosphodiester bonds**, forming a continuous mRNA chain.
8. The new RNA strand is synthesised in the **5' to 3' direction**.
9. As RNA polymerase moves along the DNA, it continues to unwind the double helix, exposing 10 to 20 DNA bases for base-pairing with RNA nucleotides.

<u>Termination of transcription</u>

10. Transcription proceeds until RNA polymerase transcribes the **polyadenylation signal sequence** (**TTATTT**) / **termination signal sequence** on the DNA
11. The RNA transcript is then cleaved **10-35 nucleotides downstream of the AAUAAA signal**, releasing the **primary RNA transcript**.
12. RNA polymerase II dissociates from the DNA and the DNA rewinds to form a double helix.
13. Primary RNA transcript need to undergo post-transcriptional modification to become mature mRNA before translation can occur.

2(b)

1. Molecular methods used in classification of organisms involve using molecular homology such as similarities in DNA or amino acid sequences;

2. molecular methods are objective and quantifiable ways to classify organisms base on their phylogenetic relationships;

3. By comparing the DNA or amino acid sequence of different organisms , we can determinate the amount of similarity in the sequences of each birds;

4. the higher the percentage of similarities in DNA or amino acid sequence, the more closely related the organisms are; Accept: vice versa

5. If the species have evolved from the same common ancestors, their DNA or amino acid sequence should show a high degree of homology with each other's and also with that of the original species;

6. based on the idea of molecular clock, different species can be arranged in chronological order of time of evolution, thus creating a phylogeny and extrapolation in phylogenetic studies;

7. Molecular methods can pick up in silent mutation thus the mutation will be reflected in the DNA sequence;

8. New species are evolved when changes in DNA sequence are accumulated over time;

<u>6 max</u>

(c)

Variation in bacteria population due to random gene mutation;
To form new genes;
Lack proof reading mechanism during DNA replication;
Novel genes can also be introduced into bacteria via transformation, transduction and conjugation;
New alleles can be incorporated into the bacteria genome via homologous recombination;
<u>4 max</u>